

# Dominique Schröder

Professor of Privacy-Enhancing Technologies at TU Wien  
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## Academic appointments

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|-------------------------------------------------------------------------------------------------|-----------------|
| Technische Universität Wien, Vienna, Austria<br>Full Professor (Univ. Prof. with tenure)        | 2024-present    |
| Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany<br>Full Professor (W3, with tenure)  | 2016-2024       |
| Saarland University, Germany<br>Professor (W2, with tenure)                                     | 08/2015-06/2016 |
| Saarland University, Germany<br>Assistant Professor (W1, tenure-track)                          | 08/2012-07/2015 |
| University of Maryland, United States of America<br>Postdoctoral Researcher under Jonathan Katz | 02/2011-06/2012 |
| Technical University of Darmstadt, Germany<br>Doctoral Student under Marc Fischlin              | 10/2006-01/2011 |

## Education

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|-----------------------------------------------------------------------------------------------------------------------|-----------------|
| Technical University Darmstadt<br>Dr. rer. nat. (with distinction) · Cryptography                                     | 10/2006-11/2010 |
| Technical University Braunschweig<br>Diplom Informatiker. (Diploma/M.Sc.) · Computer Science · Identification Schemes | 10/2002-09/2006 |

## Research interests

How can we compute on private data and share control of digital systems without trusting a single party?

I design cryptographic protocols that distribute trust and protect sensitive information. My research asks how their security guarantees hold up when participants are compromised or protocols become part of larger systems.

## Honors and awards

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| Feodor Lynen Award<br>Humboldt Foundation                                                                                                                     | 2021 |
| Efficient Unlinkable Sanitizable Signatures from Signatures with Re-Randomizable Keys<br>Distinguished Paper · Invited from PKC 2016 to IET IFS Special Issue | 2017 |
| Delegtable Functional Signatures<br>Distinguished Paper · Invited from PKC 2016 to IET IFS Special Issue                                                      | 2017 |

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| (Efficient) Universally Composable Oblivious Transfer Using a Minimal Number of Stateless Tokens<br>Distinguished Paper · Invited from TCC 2014 to Journal of Cryptology    | 2014 |
| Intel Early Career Faculty Award<br>Intel · Awarded to 6 people in Europe (\$35,000)                                                                                        | 2013 |
| Postdoctoral Fellowship<br>Deutscher Akademischer Austausch Dienst (DAAD) · University of Maryland                                                                          | 2013 |
| Dissertation<br>TU Darmstadt · summa cum laude (with highest honors)                                                                                                        | 2012 |
| Research Fellow<br>Deutscher Akademischer Austausch Dienst (DAAD) · Research visit at the University of Maryland                                                            | 2010 |
| Best Teaching Assistant Award<br>TU Darmstadt · Lecture Introduction to Modern Cryptography                                                                                 | 2007 |
| Programming Contest N@tkids<br>Winner of the programming contest “N@tkids” · akte 2000 and ricardo.de in the third category (5000 DM).<br>Invited to the TV show akte 2000. | 2000 |

### Research visits

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|-------------------------------------------------------------------------------------|-----------------|
| Visiting Professor · University of Maryland, USA<br>Host: Prof. Katz                | 07/2023-09/2023 |
| Visiting Professor · University of Berkeley, USA<br>Host: Prof. Garg                | 08/2016         |
| Visiting Professor · Chinese University of Hong Kong, Hong Kong<br>Host: Prof. Chow | 03/2016         |
| Visiting Professor · University of Maryland, USA<br>Host: Prof. Katz                | 07/2015-08/2015 |
| Visiting Professor · University of Maryland, USA<br>Host: Prof. Katz                | 08/2013-10/2013 |
| Research Internship · University of Maryland, USA<br>Host: Prof. Katz               | 07/2010-10/2010 |

### Research funding and projects

|                                                                                             |                 |
|---------------------------------------------------------------------------------------------|-----------------|
| Unrestricted Gift<br>Google · sole PI · 120.000 EUR                                         | 10/2026-12/2028 |
| Foundations and Applications of Resource-Restricted Cryptography<br>WWTF · PI · 299.708 EUR | 01/2026-12/2029 |
| Confidential 6G<br>EU Horizon · PI · 282.500 EUR                                            | 01/2023-12/2026 |
| Post-quantum Secure Key-Exchange<br>AUDI · Research Contract · 100.000 EUR                  | 04/2022-03/2023 |
| Password-based Cryptography<br>DFG · sole PI · 327.000 EUR                                  | 04/2021-03/2023 |

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| NCT: Privacy Enhancing Technologie<br>State of Bavaria · PI · 351.000 EUR                                   | 10/2021-09/2025 |
| Privacy-Preserving Blockchains<br>Protocol Labs · PI · 23.000 EUR                                           | 10/2023-09/2024 |
| Postquantum Security for Automobiles<br>AUDI · Research Contract · 60.000 EUR                               | 01/2021-12/2021 |
| Cybercrime and Forensic Computing<br>DFG Integrated Research Training Group · PI · 645.000 EUR              | 08/2019-07/2028 |
| PROMISE - Personal Medial Safe<br>BMBF · Leading PI · 710.000 EUR                                           | 01/2016-12/2019 |
| Password-based Cryptography<br>DAAD · PI · 14.000 EUR                                                       | 01/2018-12/2019 |
| Methods and Tools for Understanding Privacy<br>DFG - Collaborative Research Center (SFB) · PI · 325.000 EUR | 01/2016-12/2019 |
| CISPA Center for IT-Security, Privacy, and Accountability<br>BMBF · PI · 690.000 EUR                        | 01/2013-04/2018 |
| Cryptographic Protocols for Cloud Storage<br>DAAD · sole PI · 12.000 EUR                                    | 01/2015-12/2016 |
| Cryptography for Secure Digital Interaction<br>EU COST · PI · 135.000 EUR                                   | 11/2013-04/2018 |
| Usable Security for Mobile Devices<br>BMBF · PI · 100.000 EUR                                               | 01/2014-12/2014 |
| TWC Small: Exploring Cryptographic Models<br>NSF · PI, but I left to Germany · 314.000 EUR                  | 09/2012-08/2015 |

## Publications

|                                                                                                                                                                         |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| BitPriv: A Privacy-Preserving Protocol for DeFi Applications on Bitcoin<br>Ioannis Alexopoulos, Zeta Avarikioti, Paul Gerhart, Matteo Maffei, and<br>Dominique Schröder | NDSS<br>2027      |
| Fully Adaptive FROST with Identifiable Aborts from AOMDL<br>Ruben Baecker, Paul Gerhart, Davide Li Calsi, Luigi Russo, Dominique Schröder,<br>and Arkady Yerukhimovich  | ASIACRYPT<br>2026 |
| Fair Distributed Exchange via Threshold Adaptor Signatures<br>Ruben Baecker, Paul Gerhart, Jonathan Katz, and Dominique Schröder                                        | ASIACRYPT<br>2026 |
| Universally Composable Adaptor Signatures<br>Paul Gerhart, Daniel Rausch, and Dominique Schröder                                                                        | ACM CCS<br>2026   |
| Fully-Adaptive Two-Round Threshold Schnorr Signatures from DDH<br>Paul Gerhart, Davide Li Calsi, Luigi Russo, and Dominique Schröder                                    | EUROCRYPT<br>2026 |
| Password-Hardened Encryption Revisited<br>Ruben Baecker, Paul Gerhart, and Dominique Schröder                                                                           | ASIACRYPT<br>2025 |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Universally Composable and Round-Optimal Password-Hardened Encryption<br>Behzad Abdolmaleki, Ruben Baecker, Paul Gerhart, Mike Graf, Mojtaba Khalili, Daniel Rausch, Fritz Schmidt, and Dominique Schröder | ASIACRYPT<br>2025                                       |
| A Fully-Adaptive Threshold Partially-Oblivious PRF<br>Ruben Baecker, Paul Gerhart, Daniel Rausch, Dominique Schröder                                                                                       | CRYPTO<br>2025                                          |
| Increasing the Resilience of Secure Multiparty Computation using Security Modules<br>Lamaya Omar, Felix Freiling, Dominique Schröder                                                                       | Transactions on Dependable and Secure Computing<br>2025 |
| Automated Analysis and Synthesis of Message Authentication Codes<br>Stefan Milius, Dominik Paulus, Julian Thomas, Dominique Schröder, and Lutz Schröder                                                    | CSF<br>2025                                             |
| SoK: Descriptive Statistics Under Local Differential Privacy<br>René Raab, Pascal Berrang, Paul Gerhart, and Dominique Schröder                                                                            | PETS<br>2025                                            |
| Measuring Conditional Anonymity - A Global Study<br>Pascal Berrand, Paul Gerhart, and Dominique Schröder                                                                                                   | PETS<br>2024                                            |
| Foundations of Adaptor Signatures<br>Paul Gerhart, Dominique Schröder, Pratik Soni, and Sri A. K. Thyagarajan                                                                                              | EUROCRYPT<br>2024                                       |
| Practical Schnorr Threshold Signatures without the Algebraic Group Model<br>Hien Chu, Paul Gerhart, Tim Ruffing, Dominique Schröder                                                                        | CRYPTO<br>2023                                          |
| ROAST: Robust Asynchronous Schnorr Threshold Signatures<br>Tim Ruffing, Viktoria Ronge, Elliott Jin, Jonas Schneider-Bensch, and Dominique Schröder                                                        | ACM CCS<br>2022                                         |
| Verifiable Timed Linkable Ring Signatures For Scalable Payments for Monero<br>Sri A. K. Thyagarajan, Giulio Malavolta, Fritz Schmidt, Dominique Schröder                                                   | ESORICS<br>2022                                         |
| Inner Product Functional Commitments with Constant-Size Public Parameters and Openings<br>Hien Chu, Dario Fiore, Dimitris Kolonelos, and Dominique Schröder                                                | SCN<br>2022                                             |
| Everlasting UC Commitments from Fully Malicious PUFs<br>Bernardo Magri, Giulio Malavolta, Dominique Schröder, and Dominique Unruh                                                                          | Journal of Cryptology<br>2022                           |
| A Security Framework for Distributed Ledgers<br>Mike Graf, Daniel Rausch, Viktoria Ronge, Christoph Egger, Ralf Küsters, and Dominique Schröder                                                            | ACM CCS<br>2021                                         |
| OpenSquare: Decentralized Repeated Modular Squaring Service<br>Sri A. K. Thyagarajan, Adithya Bhat, Tiantian Gong, Aniket Kate, and Dominique Schröder.                                                    | ACM CCS<br>2021                                         |

- CoinJoin in the Wild - An Empirical Analysis in Dash** ESORICS  
2021  
Dominic Deuber and Dominique Schröder
- Foundations of Ring Sampling** PETS  
2021  
Viktoria Ronge, Christoph Egger, Russell W. F. Lai, Dominique Schröder,  
Hoover H. F. Yin
- Controlling My Genome With My Smartphone a First Clinical Experiences  
Of The PROMISE-System** Clinical Research in  
Cardiology  
2021  
Ali Amr, Marc Hinderer, Lena Griebel, Dominic Deuber, Christoph Egger,  
Farbod Sedaghat-Hamedani, Elham Kayvanpour, Daniel Huhn, Jan Haas, Karen  
Frese, Marc Schweig, Ninja Marnau, Annika Krämer, Claudia Durand, Florian  
Battke, Hans-Ulrich Prokosch, Michael Backes, Andreas Keller, Dominique  
Schröder, Hugo A. Katus, Norbert Frey, Benjamin Meder
- Threshold Password-Hardened Encryption Services** ACM CCS  
2020  
Julian Brost, Christoph Egger, Russell W. F. Lai, Fritz Schmid, Dominique  
Schröder, and Markus Zoppelt  
pp. 409-424
- Verifiable Timed Signatures Made Practical** ACM CCS  
2020  
Sri A. K. Thyagarajan, Adithya Bhat, Giulio Malavolta, Nico Döttling, Aniket Kate,  
and Dominique Schröder  
pp. 1733-1750
- The Patient as Genomic Data Manager - Evaluation of the PROMISE App.** MIE  
2020  
Lena Griebel, Marc Hinderer, Ali Amr, Benjamin Meder, Marc Schweig, Dominic  
Deuber, Christoph Egger, Claudia Kawohl, Annika Krämer, Isabell Flade,  
Dominique Schröder, Hans-Ulrich Prokosch  
pp. 1061-1065
- Feasibility and Infeasibility of Secure Computation with Malicious PUFs** Journal of Cryptology  
2020  
Dana Dachman-Soled, Nils Fleischhacker, Jonathan Katz, Anna Lysyanskaya,  
and Dominique Schröder  
pp. 595-617
- Omniring: Scaling Up Private Payments Without Trusted Setup - Formal  
Foundations and Constructions of Ring Confidential Transactions with  
Log-size Proofs** ACM CCS  
2019  
Russell W. F. Lai, Viktoria Ronge, Tim Ruffing, Dominique Schröder, Sri A. K.  
Thyagarajan, and J.Wang  
pp. 31-48
- Efficient Invisible and Unlinkable Sanitizable Signatures** PKC  
2019  
Xavier Bultel, Pascal Lafourcade, Rusell W.F. Lai, Giulio Malavolta, Dominique  
Schröder, and Sri A. K. Thyagarajan  
pp. 159-189
- My Genome Belongs to Me: Controlling Third Party Computation on  
Genomic Data** PETS  
2019  
Dominic Deuber, Christoph Egger, Katharina Fech, Giulio Malavolta, Sri A. K.  
Thyagarajan, Florian Battke, Claudia Durand, and Dominique Schröder  
pp. 108-132

- Tight Security Proofs for Schnorr Signatures**  
Nils Fleischhacker, Tibor Jager, and Dominique Schröder  
pp. 566-599  
*Journal of Cryptology* 2019
- (Efficient) Universally Composable Oblivious Transfer Using a Minimal Number of Stateless Tokens**  
Seung Geol Choi, Jonathan Katz, Dominique Schröder, Arkady Yerukhimovich, Hong-Sheng Zhou  
pp. 459-497  
*Journal of Cryptology* 2019
- Group ORAM for privacy and access control in outsourced personal records**  
Matteo Maffei, Giulio Malavolta, Manuel Reinert, Dominique Schröder  
pp. 1-47  
*Journal of Computer Security* 2019
- Homomorphic Secret Sharing for Low Degree Polynomials**  
Russell W. F. Lai, Giulio Malavolta, and Dominique Schröder  
pp. 279-309  
ASIACRYPT 2018
- Simple Password-Hardened Encryption Services**  
Russell W. F. Lai, Christoph Egger, Manuel Reinert, Sherman S. M. Chow, Matteo Maffei, and Dominique Schröder  
pp. 1405-1421  
USENIX 2018
- Functional Credentials**  
Dominic Deuber, Matteo Maffei, Giulio Malavolta, Max Rabkin, Dominique Schröder, Mark Simkin.  
pp. 64-84  
PETS 2018
- Burning Zerocoins for Fun and for Profit: A Cryptographic Denial-of-Spending Attack on the Zerocoin Protocol**  
Tim Ruffing, Sri A. K. Thyagarajan, Viktoria Ronge, Dominique Schröder  
pp. 116-119  
Crypto Valley Conference on Blockchain Technology 2018
- Subset Predicate Encryption and Its Applications**  
Jonathan Katz, Matteo Maffei, Giulio Malavolta, and Dominique Schröder  
CANS 2017
- On the Security of Frequency-Hiding Order-Preserving Encryption**  
Matteo Maffei, Manuel Reinert, and Dominique Schröder  
pp. 115-134  
CANS 2017
- Efficient Ring Signatures in the Standard Model**  
Giulio Malavolta and Dominique Schröder  
pp. 128-157  
ASIACRYPT 2017
- Phoenix: Rebirth of a Cryptographic Password-Hardening Service**  
Russell W.F.Lai, Christoph Egger, Dominique Schröder, and Sherman S.M.Chow  
pp. 899-916  
USENIX 2017
- Maliciously Secure Multi-Client ORAM**  
Matteo Maffei, Giulio Malavolta, Manuel Reinert, and Dominique Schröder  
pp. 645-664  
ACNS 2017

- Security of Blind Signatures Revisited  
Dominique Schröder and Dominique Unruh  
pp. 470–494  
Journal of Cryptology  
2017
- Efficient Cryptographic Password Hardening Services From Partially Oblivious Commitments  
Jonas Schneider, Nils Fleischhacker, Dominique Schröder, and Michael Backes  
pp. 1192–1203  
ACM CCS  
2016
- Efficient Sanitizable Signatures without Random Oracles  
Russell W. F. Lai, Tao Zhang, ShermanChow and Dominique Schröder  
pp. 363–380  
ESORICS  
2016
- Two Message Oblivious Evaluation of Cryptographic Functionalities  
Nico Doettling, Nils Fleischhacker, Johannes Krupp, and Dominique Schröder  
pp. 619–648  
CRYPTO  
2016
- Delegtable Functional Signatures  
Michael Backes, Sebastian Meiser, and Dominique Schröder  
pp. 357–386  
PKC  
2016
- Efficient Unlinkable Sanitizable Signatures from Signatures with Re-Randomizable Keys  
Nils Fleischhacker, Johannes Krupp, Giulio Malavolta, Jonas Schneider, Dominique Schröder, and Mark Simkin  
pp. 301–330  
PKC  
2016
- Nearly Optimal Verifiable Data Streaming  
Johannes Krupp, Dominique Schröder, Mark Simkin, Dario Fiore, Giuseppe Ateniese, and Stefan Nuernberger  
pp. 417–445  
PKC  
2016
- Efficient Unlinkable Sanitizable Signatures from Signatures with Re-Randomizable Keys  
Nils Fleischhacker, Giulio Malavolta, Dominique Schröder, and Mark Simkin  
pp. 166–183  
IET IFS Special Issue  
2016
- Delegtable Functional Signatures  
Sebastian Meiser and Dominique Schröder.  
pp. 194–206  
IET IFS Special Issue  
2016
- Efficient Pseudorandom Functions via On-the-Fly Adaptation  
Nico Döttling and Dominique Schröder  
pp. 329–350  
CRYPTO  
2015
- Privacy and Access Control for Outsourced Personal Records  
Matteo Maffei, Giulio Malavolta, Manuel Reinert, and Dominique Schröder  
pp. 341–358  
IEEE Symposium on Security and Privacy  
2015
- Liar, Liar, Coins on Fire! Penalizing Equivocation By Loss of Bitcoins  
Tim Ruffing, Aniket Kate, and Dominique Schröder  
pp. 219–230  
ACM CCS  
2015

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Verifiably Encrypted Signatures: Security Revisited and a New Construction<br>Christian Hanser, Max Rabkin, and Dominique Schröder                                                                           | ESORICS<br>2015   |
| VeriStream – A Framework for Verifiable Data Streaming<br>Dominique Schröder and Mark Simkin<br>pp. 146-164                                                                                                  | FC<br>2015        |
| Foundations of Reconfigurable PUFs<br>Jonas Schneider and Dominique Schröder<br>pp. 579-594                                                                                                                  | ACNS<br>2015      |
| Feasibility and Infeasibility of Secure Computation with Malicious PUFs<br>Dana Dachman-Soled, Nils Fleischhacker, Jonathan Katz, Anna Lysyanskaya, and Dominique Schröder<br>pp. 405-420                    | CRYPTO<br>2014    |
| Tight Security Proofs for Schnorr Signatures<br>Nils Fleischhacker, Tibor Jager, and Dominique Schröder<br>pp. 512-531                                                                                       | ASIACRYPT<br>2014 |
| (Efficient) Universally Composable Oblivious Transfer Using a Minimal Number of Stateless Tokens<br>Seung Geol Choi, Jonathan Katz, Dominique Schröder, Arkady Yerukhimovich, Hong-Sheng Zhou<br>pp. 638-662 | TCC<br>2014       |
| Brief Announcement: Towards Security and Privacy for Outsourced Data in the Multi-Party Setting<br>Matteo Maffei, Giulio Malavolta, Manuel Reinert, and Dominique Schröder<br>pp. 144-146                    | PODC<br>2014      |
| Demo: Security and Privacy with Google Glass<br>Johannes Krupp, Dominique Schröder, and Mark Simkin<br>pp. 1445-1447                                                                                         | CCS<br>2014       |
| Ubic: Bridging the gap Between Digital Cryptography and the Physical World<br>Mark Simkin, Dominique Schröder, Andreas Bulling, and Mario Fritz<br>pp. 56-75                                                 | ESORICS<br>2014   |
| WebTrust – A Comprehensive Authenticity and Integrity Framework for HTTP<br>Michael Backes, Reiner W. Gerling, Sebastian Gerling, Stefan Nürnberger, Dominique Schröder, and Mark Simkin<br>pp. 401-418      | ACNS<br>2014      |
| Verifiable Data Streaming<br>Dominique Schröder and Heike Schröder<br>pp. 953-964                                                                                                                            | CCS<br>2012       |



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| Uniqueness is a Different Story: Impossibility of Verifiable Random Functions from Trapdoor Permutations<br>Dario Fiore and Dominique Schröder<br>pp. 636-653 | TCC<br>2012                                |
| Security of Blind Signatures Revisited<br>Dominique Schröder and Dominique Unruh<br>pp. 662-679                                                               | PKC<br>2012                                |
| History-Free Sequential Aggregate Signatures<br>Marc Fischlin, Anja Lehmann, and Dominique Schröder<br>pp. 113-130                                            | SCN<br>2012                                |
| Security of Blind Signatures Under Aborts and Applications to Adaptive Oblivious Transfer<br>Marc Fischlin and Dominique Schröder.<br>pp. 169-204             | Journal Mathematical<br>Cryptology<br>2012 |
| Round Optimal Blind Signatures<br>Dominique Schröder and Dominique Unruh<br>pp. 630-648                                                                       | CRYPTO<br>2011                             |
| Impossibility of Blind Signatures from One-Way Permutations<br>Jonathan Katz, Dominique Schröder, Arkady Yerukhimovich<br>pp. 615-629                         | TCC<br>2011                                |
| How to Aggregate the CL Signature<br>Dominique Schröder<br>pp. 298-314                                                                                        | ESORICS<br>2011                            |
| Expedient Non-Malleability Notions for Hash Functions<br>Paul Baecher, Marc Fischlin, and Dominique Schröder<br>pp. 268-283                                   | CT-RSA<br>2011                             |
| On the Impossibility of Three-Move Blind Signature Schemes<br>Marc Fischlin and Dominique Schröder<br>pp. 197-215                                             | EUROCRYPT<br>2010                          |
| Unlinkability of Sanitizable Signatures<br>Christina Brzuska, Marc Fischlin, Anja Lehmann, and Dominique Schröder<br>pp. 444-461                              | PKC<br>2010                                |
| Confidential Signatures and Deterministic Signcryption<br>Alex W. Dent, Marc Fischlin, Mark Manulis, Dominique Schröder, and Martijn Stam<br>pp. 462-479      | PKC<br>2010                                |

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| <p><b>Redactable Signatures for Tree-Structured Data: Definitions and Construction</b></p> <p>Christina Brzuska, Heike Busch, Özgür Dagdelen, Marc Fischlin, Martin Franz, Stefan Katzenbeisser, Mark Manulis, Cristina Onete, Andreas Peters, Bertam Poettering, and Dominique Schröder</p> <p>pp. 87-104</p> | <p>ACNS<br/>2010</p>        |
| <p><b>Generic Constructions for Verifiably Encrypted Signatures without Random Oracles or NIZKs</b></p> <p>Markus Rückert, Michael Schneider, and Dominique Schröder</p> <p>pp. 69-86</p>                                                                                                                      | <p>ACNS<br/>2010</p>        |
| <p><b>History-Free Aggregate Message Authentication Codes</b></p> <p>Oliver Eikemeier, Marc Fischlin, Jens-Fabian Götzmann, Anja Lehmann, Dominique Schröder, Peter Schröder, and Daniel Wagner</p> <p>pp. 309-328</p>                                                                                         | <p>SCN<br/>2010</p>         |
| <p><b>Fair Partially Blind Signatures</b></p> <p>Markus Rückert and Dominique Schröder</p> <p>pp. 34-51</p>                                                                                                                                                                                                    | <p>Africacrypt<br/>2010</p> |
| <p><b>Public-Key Encryption with Non-interactive Opening: New Constructions and Stronger Definitions</b></p> <p>David Galindo, Benoit Libert, Marc Fischlin, Georg Fuchsbauer, Anja Lehmann, Mark Manulis, and Dominique Schröder</p> <p>pp. 333-350</p>                                                       | <p>Africacrypt<br/>2010</p> |
| <p><b>CAPTCHAs: The Good, the Bad, and the Ugly.</b></p> <p>Paul Baecher, Lior Gordon, Marc Fischlin, Michael Luetzow, Dominique Schröder</p> <p>pp. 353-365</p>                                                                                                                                               | <p>Sicherheit<br/>2010</p>  |
| <p><b>Security of Blind Signatures Under Aborts</b></p> <p>Marc Fischlin and Dominique Schröder</p> <p>pp. 117-128</p>                                                                                                                                                                                         | <p>PKC<br/>2009</p>         |
| <p><b>Security of Sanitizable Signatures Revisited</b></p> <p>Christina Brzuska, Marc Fischlin, Tobias Freudenreich, Anja Lehmann, Marcus Page, Jakob Schelbert, Dominique Schröder, and Florian Volk</p> <p>pp. 297-316</p>                                                                                   | <p>PKC<br/>2009</p>         |
| <p><b>Security of Verifiably Encrypted Signatures and a Construction Without Random Oracles</b></p> <p>Markus Rückert and Dominique Schröder</p> <p>pp. 17-34</p>                                                                                                                                              | <p>Pairings<br/>2009</p>    |
| <p><b>Aggregate and Verifiably Encrypted Signatures from Multilinear Maps without Random Oracles</b></p> <p>Markus Rückert and Dominique Schröder</p> <p>pp. 750-759</p>                                                                                                                                       | <p>ISA<br/>2009</p>         |

Sanitizable Signatures: How to Partially Delegate Control for  
Authenticated Data BioSig  
2009  
Christina Brzuska, Marc Fischlin, Anja Lehmann, and Dominique Schröder  
pp. 117-128

Mobi: Eine Infrastruktur fuer das Internet der Dinge Informatiktage  
2006  
Michael Döring and Dominique Schröder

## Patents

Methods for threshold password-hardened encryption and decryption 2022  
Russell W.F. Lai, Dominique Schröder, Christoph Egger, Julian Brost · US · 11323248

## Doctoral supervision

Davide Li Calsi 2025 - present  
Distributed Systems  
Ongoing

Ruben Baecker 2023 - present  
Privacy-Preserving Access Control  
Ongoing

Julian Thomas 2023 - present  
Formalizations of privacy-enhancing cryptographic primitives  
Ongoing

Paul Gerhart 2022 - present  
Fair exchange, optimized Blockchain protocols and privacy  
Ongoing

Hien Chu 2021 - present  
Privacy-preserving techniques, threshold cryptography  
Ongoing

Victoria Ronge 2018 - 2023  
Security and Privacy of Cryptocurrency Signatures  
First position after PhD: Industry  
Current position: Industry

Dominic Deuber 2017 - 2023  
Cryptocurrencies Within The Conflicting Tension Between Law and IT Security  
First position after PhD: Postdoc at FAU  
Current position: Industry

Russel W. F. Lai 2017 - 2022  
Succinct Arguments: Constructions and Applications  
First position after PhD: Assistant Professor at Aalto University  
Current position: Assistant Professor at Aalto University

Christoph Egger 2016 - 2022  
On Abstraction and Modularization in Protocol Analysis  
First position after PhD: Researcher at Institut de Recherche en Inf. Fondamentale  
Current position: Assistant Professor at Chalmers University of Technology

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| <b>Sri Aravinda Krishnan Thyagarajan</b><br>Cryptographic Locks for Scriptless Cryptocurrency Payments<br>First position after PhD: Postdoc at Carnegie Mellon University<br>Current position: Assistant Professor at the University of Sydney | 2016 - 2021 |
| <b>Guilio Malavolta</b><br>Cryptographic Clocks and Applications<br>First position after PhD: UC Berkeley and Carnegie Mellon University<br>Current position: Assistant Professor Bocconi University                                           | 2013 - 2019 |
| <b>Oliver Willers</b><br>MEMS Sensors As Physical Unclonable Functions<br>First position after PhD: Bosch<br>Current position: Bosch                                                                                                           | 2015 - 2019 |
| <b>Tim Ruffing</b><br>Cryptography for Bitcoin and Friends<br>First position after PhD: Blockstream<br>Current position: Blockstream                                                                                                           | 2014 - 2019 |
| <b>Nils Fleischhacker</b><br>Minimal Assumptions in Cryptography<br>First position after PhD: Postdoc at Carnegie Mellon University and Johns Hopkins University<br>Current position: Assistant Professor at Ruhr University Bochum            | 2012 - 2016 |

## Postdoctoral mentoring

|                                                                                                        |                   |
|--------------------------------------------------------------------------------------------------------|-------------------|
| <b>Luigi Russo</b>                                                                                     | 07/2025 - present |
| <b>Dominic Deuber</b><br>Subsequent position: IT security expert at the German Accreditation Body      | 03/2023 - 06/2024 |
| <b>Mojtaba Khalili</b><br>Subsequent position: Assistant Professor, Isfahan University of Technology   | 09/2022 - 09/2023 |
| <b>Albrecht Petzoldt</b><br>Subsequent position: Dagstuhl                                              | 09/2019 - 08/2022 |
| <b>Bernardo Magri</b><br>Subsequent position: Associate Professor at the University of Manchester, UK. | 11/2016 - 11/2018 |

## Teaching

|                                                                 |                  |
|-----------------------------------------------------------------|------------------|
| <b>Advanced Privacy Enhancing Technologies</b><br>TUW · BSC/MSC | 2025             |
| <b>Data Privacy</b><br>FAU · BSC                                | 2025             |
| <b>IT Security &amp; Law</b><br>FAU · Master                    | 2025, 2024, 2023 |
| <b>Privacy Enhancing Technologies</b><br>TUW · BSC              | 2025             |
| <b>Blockchain und Kryptowährungen</b><br>FAU · BSC              | 2024, 2022, 2020 |

|                                                                     |                                             |
|---------------------------------------------------------------------|---------------------------------------------|
| Data Privacy<br>FAU · BSC/MSC                                       | 2024, 2023, 2022                            |
| Introduction to Algorithms<br>FAU · BSC                             | 2024, 2023                                  |
| Introduction to Data Privacy<br>FAU · BSC/MSC                       | 2023, 2021                                  |
| Introduction to Modern Cryptography<br>FAU · BSC/MSC                | 2023, 2022, 2021, 2020,<br>2018, 2017, 2016 |
| Privacy and Legal Aspects and Their Realizations<br>FAU · BSC/MSC   | 2022, 2021                                  |
| Privacy-Preserving Cryptocurrencies<br>FAU · BSC/MSC                | 2022, 2021, 2020                            |
| Secure Multi-Party Computation<br>FAU · BSC/MSC                     | 2021, 2017                                  |
| Foundations of Privacy<br>FAU · BSC/MSC                             | 2020                                        |
| Password-based Cryptography<br>FAU · BSC/MSC                        | 2020, 2019, 2018                            |
| Research Oriented Cryptography<br>UdS · BSC/MSC                     | 2020, 2014                                  |
| Cryptocurrencies<br>FAU · BSC/MSC                                   | 2019, 2018, 2016                            |
| Sabbatical<br>FAU · -                                               | 2019                                        |
| Algorithms and Data Structures<br>FAU · BSC                         | 2018                                        |
| Foundations of Cryptocurrencies<br>FAU · BSC/MSC                    | 2017                                        |
| Public-Key Encryption<br>UdS · BSC/MSC                              | 2017, 2015, 2014, 2013, 2012                |
| Cryptocurrencies<br>UdS · BSC/MSC                                   | 2016                                        |
| Secure Multi-Party Computation<br>UdS · BSC/MSC                     | 2016, 2013                                  |
| Introduction to Modern Cryptography<br>UdS · BSC/MSC                | 2015, 2014                                  |
| Recent Advances in Secure Multi-Party Computations<br>UdS · BSC/MSC | 2015                                        |
| <b>Conference leadership</b>                                        |                                             |
| ACM CCS Area Chair Applied Cryptography                             | 2025, 2026                                  |

|                                                                                      |      |
|--------------------------------------------------------------------------------------|------|
| Chair of IEEE International Conference on Blockchain                                 | 2023 |
| Co-Chair of IEEE TEMS on Blockchain and Distributed Ledger Technologies.             | 2021 |
| General Chair of IACR Theory of Cryptography Conference (TCC).                       | 2019 |
| Poster Session Chair of ACM Conference on Computer and Communications Security (CCS) | 2016 |

### Editorial boards

|                                          |           |
|------------------------------------------|-----------|
| IACR Communications in Cryptology        | 2024-2026 |
| ACM Transactions on Privacy and Security | 2024      |

### Program committees (selected)

|                                                                |                        |
|----------------------------------------------------------------|------------------------|
| USENIX Security                                                | 2026                   |
| ACM Conference on Computer and Communications Security         | 2019, 2023, 2024-2026  |
| Advances in Cryptology (CRYPTO)                                | 2016, 2019, 2020, 2023 |
| Advances in Cryptology (EUROCRYPT)                             | 2015, 2023, 2027       |
| IEEE Computer Security Foundations Symposium                   | 2020-2024              |
| Advances in Cryptology (ASIACRYPT)                             | 2022                   |
| Privacy Enhancing Technologies Symposium (PETS)                | 2021, 2022, 2024-2027  |
| Public-Key Cryptography (PKC)                                  | 2023-2025              |
| The Cryptographers' Track at RSA Conference (CT-RSA)           | 2025                   |
| European Symp. on Research in Computer Security (ESORICS)      | 2021-2026              |
| Conf. on Cryptology and Network Security (CANS)                | 2016-2019, 2021-2023   |
| IEEE Security and Privacy (Oakland)                            | 2020, 2021             |
| Advances in Cryptology (INDOCRYPT)                             | 2021                   |
| CODASPY                                                        | 2024                   |
| IEEE European Symposium on Security and Privacy (EuroSP)       | 2018                   |
| Asia Conference on Computer and Comm. Security (ASIACCS)       | 2017, 2018             |
| Conference on Security and Cryptography for Networks (SCN)     | 2018, 2022             |
| Conference on Applied Cryptography and Network Security (ACNS) | 2024                   |
| International Conference on Provable Security (PROVSEC)        | 2016                   |
| ACM Conference on Pervasive and Ubiquitous Computing (UbiComp) | 2016                   |

### Grant reviewing

|                                       |
|---------------------------------------|
| European Research Council (ERC)       |
| Deutsche Forschungsgemeinschaft (DFG) |

Deutscher Akademischer Austauschdienst (DAAD)

Studienstiftung des Deutschen Volkes

Schloss Dagstuhl

Österreichische Forschungsförderungsgesellschaft (FFG)

Natural Sciences and Engineering Research Council of Canada (NSERC)

Dutch Research Council (NWO)

### Journal reviewing

Journal of Cryptology

Journal of Computer Security

IET Journal for Information Security

Journal of Mathematical Cryptology

ACM Transactions on Privacy and Security (formerly TISSEC)

IEEE Transactions on Security and Dependable Computing

### Doctoral committees

|                                                                             |      |
|-----------------------------------------------------------------------------|------|
| Jenny Ottmann                                                               | 2024 |
| 3rd reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg            |      |
| Swaroop Ready                                                               | 2024 |
| 2nd reviewer · Indian Institute of Technology, Hyderabad                    |      |
| Victoria Ronge                                                              | 2023 |
| advisor · Friedrich-Alexander Universität Erlangen-Nürnberg                 |      |
| Dominic Deuber                                                              | 2023 |
| advisor · Friedrich-Alexander Universität Erlangen-Nürnberg                 |      |
| Franz-Josef Streit                                                          | 2022 |
| chair of the commission · Friedrich-Alexander Universität Erlangen-Nürnberg |      |
| Christoph Egger                                                             | 2022 |
| advisor · Friedrich-Alexander Universität Erlangen-Nürnberg                 |      |
| Russell W. F. Lai                                                           | 2022 |
| advisor · Friedrich-Alexander Universität Erlangen-Nürnberg                 |      |
| Sri Aravinda Krishnan Thyagarajan                                           | 2021 |
| advisor · Friedrich-Alexander Universität Erlangen-Nürnberg                 |      |
| Markus Zoppelt                                                              | 2021 |
| 1st reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg            |      |
| Katja Auernhammer                                                           | 2021 |
| 3rd reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg            |      |
| Kristof Teichel                                                             | 2021 |
| 2nd reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg            |      |

|                                                                                                  |      |
|--------------------------------------------------------------------------------------------------|------|
| Daniel Rausch<br>2nd Reviewer · Universität Stuttgart                                            | 2020 |
| Guilio Malavolta<br>advisor · Friedrich-Alexander Universität Erlangen-Nürnberg                  | 2019 |
| Oliver Willers<br>1st reviewer, co-advisor · Saarland University                                 | 2019 |
| Tim Ruffing<br>co-advisor · Saarland University                                                  | 2019 |
| Andrea Zimmermann<br>external examiner · Friedrich-Alexander Universität Erlangen-Nürnberg       | 2018 |
| Nils Fleischhacker<br>advisor · Saarland University                                              | 2017 |
| Yan Zhuang<br>chair of the commission · Friedrich-Alexander Universität Erlangen-Nürnberg        | 2017 |
| Mykolai Protsenko<br>chair of the commission · Friedrich-Alexander Universität Erlangen-Nürnberg | 2017 |
| Thomas P. Jakobsen<br>2nd reviewer · Aarhus University, Denmark                                  | 2015 |
| Esfandiar Mohammadi<br>2nd reviewer · Saarland University                                        | 2014 |

### Administrative service

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| Head of 6 and member of 11 hiring committees                           | Since 2017 |
| FAU Study commission B. Sc. computer/IT-security                       | since 2017 |
| Department of computer science board of management FAU                 | 2016-2024  |
| FAU Faculty council of the faculty of engineering (re-elected in 2021) | 2019-2024  |
| FAU Department Informatik   speaker                                    | 2023-2024  |
| FAU Department Informatik   Deputy speaker                             | 2021-2023  |
| Study commission computer science                                      | 2018-2021  |
| Scientific steering committee of the NCT                               | 2016-2017  |
| Master admission committee (Saarland University)                       | 2012-2016  |
| development of the Bachelor “Cybersicherheit” (Saarland University)    | 2014       |

### Expert contributions and public service

|                                                                                                               |      |
|---------------------------------------------------------------------------------------------------------------|------|
| Deutscher Bundestag · Health Committee<br>Independent expert contribution: Privacy and the use of health data | 2023 |
| Deutscher Bundestag · Health Committee<br>Independent expert contribution: Patient Data Protection Act (PDSG) | 2020 |
| Deutscher Bundestag · Health Committee<br>Independent expert contribution: Digital Healthcare Act (DVG)       | 2019 |



## Selected talks and appearances

|                                                                                                                                   |      |
|-----------------------------------------------------------------------------------------------------------------------------------|------|
| Privacy of Medical Data<br>Humboldt Foundation, Network Meeting, Mainz                                                            | 2023 |
| Welchen Beitrag leistet die Digitalisierung für eine gerechte<br>Gesundheitsversorgung in Deutschland?<br>Leopoldina, Berlin, DE. | 2021 |
| Data Privacy for Medical Records<br>TU Graz, Austria.                                                                             | 2019 |
| Privacy and Access Control for Outsourced Personal Records.<br>Leopoldina, Saarbrücken, DE.                                       | 2016 |
| Efficient Pseudorandom Functions via On-the-Fly Adaptation.<br>Chinese University of Hong Kong (CUHK), Hong Kong.                 | 2015 |
| Bridging the Gap Between Digital Cryptography and the Physical World.<br>ATM Security, London, UK.                                | 2014 |
| Ubicrypt - Die Unterschrift der Zukunft.<br>Future Talk, CeBiT, Hannover, Germany.                                                | 2014 |
| Security and Privacy in the Internet.<br>SR Television. Interview with Joachim Weyand                                             | 2013 |
| Verifiable Data Streaming<br>Intel Collaborative Research Institute for Secure Computing, Darmstadt, Germany.                     | 2013 |
| Verifiable Data Streaming<br>University of Maryland, USA.                                                                         | 2013 |
| Verifiable Data Streaming<br>Brown University, USA.                                                                               | 2013 |
| Verifiable Data Streaming<br>IMDEA, Madrid Institute of Advanced Studies, Madrid, Spain.                                          | 2013 |
| Provable Security and Beyond.<br>Florida State University, Tallahassee, USA.                                                      | 2012 |
| History-Free Aggregate Message Authentication Codes.<br>Universite de Rennes, Rennes, France.                                     | 2010 |